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The Art of Children's Drawing

ELIZABETH ROSENBLATT and ELLEN WINNER

Prior to the latter half of the nineteenth century, children's drawings were seen primarily as unskilled and primitive attempts at representation and hence received no serious attention. But with the growth of interest in child development at the end of the nineteenth century came a surge of interest in children's drawings. Perhaps because children's drawings look so strikingly different at different ages, drawings were seen as providing a sharply focussed lens through which to view development.

Psychologists have typically viewed children's drawings as windows on personality and affect, on the one hand, or as perspectives on cognition and intelligence, on the other. Altschuler and Hattwick (1947) used drawings as indications of personality structure; Goodenough (1926) used drawings as measures of intelligence; Piaget (1963) used drawings as reflections of the child's concepts; and recently Freeman (1980) and Goodnow (1977) have made use of drawings to reveal the child's cognitive strategies such as planning and sequencing.

These approaches have often used early drawings as indications of *deficiencies* of some sort: for instance, "tadpole" drawings of humans have been seen as a reflection of a global rather than differentiated concept of the human body (Piaget, 1963) or as evidence for poor memory for body parts (Freeman, 1980); lack of detail in drawings has been taken as evidence for low intelligence (Goodenough, 1926); transparencies (when the child reveals the insides of objects drawn) have been taken as evidence that the child can only draw what he knows but not what he sees (Piaget & Inhelder, 1967); and the overuse of dark colors has been taken as evidence of psychopathology (Altschuler & Hattwick, 1947).

Only rarely have children's drawings been studied on their own terms, for what they can tell us about the child's knowledge of the symbol system of drawing and for what they can tell us about the child's aesthetic goals. Researchers in this tradition have reminded us that children's drawings may look odd not because the child lacks some set of skills possessed

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by the adult, but rather because the child's goals differ from those of the adult. The child is seen as guided less by the goal of realism than by the goal of simplicity, lack of ambiguity, and visual logic (Arnheim, 1974; Golomb, 1981; Schaeffer-Simmern, 1948). Moreover, rather than using drawings as windows onto something other than drawing, researchers in this tradition have shown us what we can learn about children's mastery of the rules specific to the medium of drawing (e.g., Arnheim, 1974; Golomb, 1981).

Research on children's drawings carried out at Project Zero has been stimulated by research in the "aesthetic" rather than the "deficiency" tradition. Our focus has been on an examination of children's drawings as a means of understanding the child's artistic knowledge, rather than as a window on some other form of knowledge. In addition, we have treated the drawings that children produce as only part of the picture. Also essential, in our view, are two other features: (1) an examination of the perceptual skills that children bring to the task of drawing; (2) a consideration of children's abilities to reflect about the kinds of processes and decisions involved in either the making or the understanding of an art object.

Our subjects have for the most part been children with no special abilities or training in drawing. Only recently have we begun to study children gifted in drawing. Our goal in working with these talented children is to determine what skills distinguish such children and whether these distinguishing abilities are apparent not only in the kinds of drawings produced, but also in the qualities perceived and the processes reflected upon.

In what follows, we focus first on studies of the making of drawings. We then turn to studies of perceptual abilities and then to a recently undertaken investigation of reflective capacities. We conclude with a description of studies of children with special abilities in the arts.

Drawings Produced

The major question that has guided our studies of drawings produced by children is the relationship between preschool drawings and those produced by twentieth-century artists on the one hand and by elementary-school-age children on the other. We have been struck by the clear resemblances between the works of very young children and those produced by contemporary artists (e.g., Klee, Miró, Picasso). Drawings by preschool children are spontaneous, fanciful, nonstereotyped, and aesthetically appealing. As children move into the middle elementary school years, they begin to draw with less frequency, and their drawings appear much more literal-minded: they are rigid and stereotyped. The child at this age seems dominated by one goal—that of drawing things the way they look or the way they look in pictures. Although their drawings are more "accurate,"

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they are considerably less pleasing aesthetically. While one might slip a painting by a five-year-old into the Museum of Modern Art, one could never get away with introducing a painting by a ten-year-old.

Hence, it appears as if something is lost with age: with increasing skill and technical competence comes a decline in aesthetic appeal. We have come to call the early years the preconventional years and the years of middle childhood the conventional years. It is only those individuals with special skill and interest in visual art who emerge from the conventional phase and begin to violate the rules mastered in the elementary school years. We call this last phase the postconventional stage. Hence the preschooler *lacks* conventions, while the adult artist may *reject* them. Because neither preschool nor adult art is thus dominated by conventions, the two types of work appear similar. In our view, the similarity between the two is more than superficial. In the intensity of involvement in drawing, in the willingness to play and experiment, in the lack of concern for "how it should be done," the preschooler resembles the adult master in nontrivial respects.

To test our view that drawings decline in aesthetic appeal and interest from the preschool to middle elementary school years, we pitted drawings by five-year-olds against ones by ten-year-olds and asked artists and nonartists which they preferred (Wan & Winner, 1980). Artists strongly preferred the preschool drawings. Adults who were not artists, however, liked each type equally. Hence, for individuals sensitized to the visual arts, children's drawings decline with age in aesthetic appeal.

It is possible that this apparent decline is due not to an actual *loss* of ability, but rather to a shift in the child's focus of concern. Perhaps ten-year-olds retain the ability to draw as imaginatively as five-year-olds, but do not choose to do so because they are striving instead for realism. To address this question, we attempted to inspire elementary-school-age children to draw once again like preschoolers (Winner et al., 1980). Ten-year-old children were first asked to make drawings of their own. They were then shown pictures that had been done by "another person" (a preschooler) and were asked to re-draw their own drawing in the way that this "other person" would have drawn it.

Ten-year-olds were able to alter their drawings in response to the instructions. However, they were only able to capture the superficial aspects of preschool drawings. Their second drawings were messier and less realistic than their first ones, but not more imaginative and flavorful. Thus, children at the conventional stage in fact may *not* be able to recapture the fanciful, playful quality of their younger drawings. It is possible, however, that these children simply did not understand what we were asking them to do—i.e., to capture the playful, imaginative quality of preschool art. Perhaps if more strongly cued they could have produced imaginative draw-

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ings. However, making playful, inventive drawings seems to be such a low priority for conventional-stage children that, given any ambiguity in the task, they fall into other patterns.

We sought further evidence for an age-related decline in a study in which children were asked to complete line drawings that were either quite realistic or sketchy and schematic (Winner et al., 1983). We wanted to determine whether conventional-stage children are willing to complete a nonrealistic picture nonrealistically or if they are unable to free themselves of the goal of realism.

Confirming our expectations, middle elementary-school-age children completed all of the drawings realistically, regardless of the level of realism in the rest of the picture. For example, a ten-year-old added identical realistic hands, drawn with all five fingers, to both a schematic Picasso drawing and a much more realistic one. In contrast, six-year-olds, while able to complete realistic drawings fairly realistically, were willing to complete schematic drawings much less realistically. For example, a six-year-old completed the unrealistic Picasso with a schematic hand—a sunlike form with a few radiating lines for fingers—but completed the realistic Picasso with a much more articulated hand. Twelve-year-olds performed like six-year-olds, completing each drawing according to the drawing's level of realism. Thus, six-year-olds performed as well as twelve-year-olds. They were able and willing to vary the level of realism according to the style of the original drawing. In contrast, eight- and ten-year-olds—children in the midst of the conventional stage—were dominated by one goal: to draw as realistically as they could regardless of the invitation provided by the picture's style.

Our studies point to the conclusion that there is indeed a real loss of aesthetic sensibility during the conventional stage. Whether the loss is inevitable or an artifact of a cultural preference for realism in art is not known. It is certainly possible that the loss is an inevitable concomitant of skill mastery. That is, children may have to re-focus their attention on rules in the initial stages of skill mastery and may simply not have the scope to be able to retain an interest in play and exploration. Perhaps only when the basic skills of drawing have become a relatively automatic part of their repertoire do they have the freedom to turn their attention back to play.

Drawings Perceived

When preschool children are asked which they prefer, a drawing by a five-year-old or one by a ten-year-old, they invariably vote for the older children's drawing. In fact, preschool children are as dismissive of their own drawings as older children are of preschool drawings (Wan & Winner,

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1980). The finding that preschoolers actually prefer pictures that are more stereotyped, realistic, and conventional is antithetical to a romantic view of the very young child as "natural, avant-garde artist." It suggests that the aesthetic effects found in their drawings may be accidental rather than intentional. Perhaps in their spontaneous drawings children strive for realism but fail to attain it.

We have tried to develop measures that can reveal the degree to which the young child is aware of, and in control of, the aesthetic effects of his drawings. We have isolated several key aesthetic properties of drawings and have examined the child's emerging sensitivity to such properties. To the extent that children show sensitivity to these properties, we can feel strengthened in our coupling of the five-year-old with the adult master. But to the extent that children are blind to these properties, we are forced to accept an antiromantic view of early art and conclude that its aesthetic appeal is more accidental than intentional.

Based on the work of Nelson Goodman and Rudolf Arnheim, research at Project Zero has focussed on children's sensitivity to the three properties of drawings that we consider to be central to the arts: *repleteness*, *expression*, and *composition*.

Repleteness refers to the "fullness" of a symbol when it is functioning as art (Goodman, 1976). For example, in a line drawing (but not in a picture functioning nonaesthetically, such as a scientific graph) one is meant to attend not only to the forms the lines create, but also to the properties of the lines themselves (e.g., the texture, thickness, rhythm). These aspects are part of the meaning of the drawing, but not the graph.

Expression refers to the metaphoric conveyance of properties (such as sadness, anxiety, loudness, sharpness) in a work of art (Goodman, 1976). These properties are conveyed through the aspects of color, line, composition, and so forth. For example, "sadness" (which may be represented through denotational symbols, such as a frowning face) may also be expressed through the use of dark colors and droopy lines.

Composition refers to the structural organization and overall balance in a work of art (Arnheim, 1974). Balanced compositions may be achieved in a number of ways. They may be achieved through symmetry, but are more commonly achieved through a dynamic, asymmetrical organization of forms. For instance, a large form may be offset by a much smaller, brighter form due to the apparent weight of brighter hues.

To the extent that children fail to attend to repleteness, expression, and composition, we must conclude that works of art are functioning for them as nonaesthetic symbols such as maps and graphs. In an initial study, we focussed on sensitivity to repleteness and expression in six-, nine-, and eleven-year-olds (Carothers & Gardner, 1979). Children were shown pairs of incomplete drawings which were identical in all respects except line

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quality or mood, respectively. The drawings were simple sketches in the style of drawings by ten-year-olds created for the purposes of this study. In the repleteness tasks, the children were asked to add a person to the picture and to draw it the way the person who drew the picture would make it. In the expression tasks, they were asked to add a tree and flower to a picture of a sad person and to a picture of a happy person. For each production task, there was an analogous multiple-choice task.

We reasoned that if children are sensitive to repleteness and expression and realize that these properties are relevant to the drawings, they should complete the picture by employing the same line quality or mood. Similarly, they should be able to select the completions which maintain the properties varied.

The findings revealed that the six-year-olds selected at random on both types of perception tasks and were also unable to complete the drawings along the dimensions required. Nine-year-olds succeeded on the perception tasks, but performed less well on the production tasks. Not until the age of eleven were children able to succeed on both types of tasks for both of the aesthetic properties.

These findings suggest that while very young children seem to have the capacity to produce quite aesthetically appealing works, they are not sensitive to certain basic aesthetic properties. Preconventional children fail to discriminate repleteness, expression, and composition; they prefer realistic pictures, yet they produce aesthetically appealing, nonrealistic pictures. In addition, at the age at which they begin to be aesthetically aware, their own drawings become highly conventionalized and therefore less pleasing. Hence, we are left with something of a mystery, about which we offer the following speculations.

Perhaps a beginning of a solution to the mystery of preconventional art comes from the results of the study in which children had to complete realistic and unrealistic pictures (Winner et al., 1983). Recall that six-year-olds were better than older children at attending to a fourth dimension—how articulated and realistic a drawing is vs. how sketchy and unrealistic. While preschool children cannot be said to manipulate repleteness, expression, and composition intentionally in the service of aesthetic creation, perhaps they are aware of the overall style of a drawing and strive to keep it consistent.

Conventional-phase children, while able to perceive such properties, seem unconcerned with them in their own drawings. It is not until early adolescence that these divergent lines of development converge, allowing both perception and production to interact. Fluidity, expressivity, and imaginativeness are now created because the adolescent *intends* to do so and is clearly aware of the effects produced.

In another line of investigation, we sought to determine whether and

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to what degree perceptual abilities in the visual arts are general skills that could be applied to any artistic domain or whether these skills are domain specific. A large-scale study of aesthetic sensitivity was designed to look at development in other artistic domains—music and literature as well as visual art (Winner et al., 1986). This allowed the examination of two questions: (1) Does perception in one domain generalize to other domains? (2) Does sensitivity to one property in a domain generalize to other properties within that same domain?

Children aged seven, nine, and twelve years were given tasks that either examined one property across all three domains (e.g., repleteness in visual art, music, and literature) or all three properties in one domain (e.g., expression, composition, and repleteness in visual art). All of the tasks involved the presentation of a target and the selection of the best completion (given two choices) and were analogous across domains. For example, for composition in visual art, children were shown an incomplete picture and two completions, one that was clearly balanced and one that was not. Then they were asked to choose the completion that finished the target "in the best way." Similarly, in music (or literature), children heard an incomplete piece of music (or story) and then two completions, and were asked to select the one that made the best ending.

Several problems with the tasks used previously (Carothers & Gardner, 1979) were addressed here. One could argue that the stimuli used in the previous study had not been produced by artists and therefore were not "art." Thus, it is possible that the children did not respond to them aesthetically, but might have if "real" works of art had been used. In addition, there was an ambiguity in the way we initially tested sensitivity to expression. In the initial task, children were asked to add a tree to both a sad and a happy picture and also to decide which of two trees (wilted vs. blooming) would best complete a picture that showed a happy person. It is possible that children saw the moods expressed by the two trees but did not realize that they should complete the pictures in the same mood. For instance, they might have deliberately added a happy tree to a sad picture. In response to these problems, all of the stimuli used in the succeeding study were taken from works by well-known artists, and the expression task was changed to a "matching" task in which children were asked to match pictures according to similarity in mood.

Correlational analyses revealed no consistent relations within *or* across the domains. That is, high performance on tests of repleteness in visual art did not predict high performance on tests of repleteness in music or literature. In addition, high performance on repleteness in visual art did not predict performance on either expression or composition in visual art. We are forced to conclude that aesthetic sensitivity is both art form specific *and* property specific. Rather than one central skill, aesthetic sensitivity

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seems to be composed of several abilities that develop along separate trajectories.

Children's Reflections about the Arts

In past studies conducted at Project Zero, reflection, a skill we believe essential to artistic learning and creation, has been ignored. Reflection entails thinking about the process of making, and the final product. The ability to reflect about one's goals, decisions, and solutions, as well as about the influences of the works of others on one's own work, seems to be crucial to cultivate in the service of any artistic endeavor.

Children's reflections about the arts were initially investigated in a study in which we interviewed children and adolescents about a broad range of topics and several aesthetic media (Gardner et al., 1975). We asked subjects to reflect about how art is made and how aesthetic judgments are made and justified. The results revealed parallels between the development of reflections about art and more general conceptions of the world (cf. Piaget, 1963). Preschool children articulated a very concrete view of art, as they do of the world in general, and were concerned primarily with the mechanics of how a work is produced. For instance, they told us that a painting is finished when the paper is filled up or when the artist gets tired. They had no notion that training of any kind is required to be an artist and even believed that animals could create art—as long as they could hold a pencil or paintbrush. By the age of seven to nine years, the child's obsession with rules spills over into the aesthetic domains: children judged the value of a drawing in terms of "neatness" and realistic rendering. Between the ages of nine to thirteen years, children began to be able to recognize the aesthetic effects that an artist has intended to create. Realistic accuracy was no longer held to be the only criterion by which works are judged. Instead, a full range of artistic properties were recognized and valued. The highly relativistic view that is characteristic of adolescence was also apparent in their judgments of the relativity of various aesthetic effects: what is good, they said, is simply a matter of opinion. In sum, reflection about the arts was shown to follow a specific course of development and to mirror trends in other areas of cognitive growth. Reflective skills were found to develop late: not until adolescence did children begin to reflect about intentional aesthetic effects of artworks, and even at this age children were unable to consider how one might justify an aesthetic judgment beyond the claim of personal preference.

Similar findings concerning the late emergence of reflective skills came from a study of children's abilities to make critical judgments in the arts (Rosenstiel et al., 1978). Children aged six, eight, eleven, and fifteen years were shown pairs of painting reproductions and were asked to decide which they preferred, which others would prefer, and which was the bet-

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ter work. The youngest children (six-year-olds) failed to distinguish among the various standards, and all of the younger age groups (six-, eight-, and eleven-year-olds) demonstrated a severely limited vocabulary with which to discuss their aesthetic judgments. Only the oldest children were able to articulate distinctions among the three standards and were able to consider a wide variety of criteria in their reflections. Their responses ranged from discussions of formal properties to the influence of specific periods in art history.

It is possible that subjects in this study made finer distinctions than they were able to verbalize. Clearly, one needs to nourish not only reflective skills, but also the ability to talk about them. A research project is now underway at Project Zero in cooperation with Princeton's Educational Testing Service and the Pittsburgh Public School System. In this study, we are attempting to address the role of reflection (and the ability to make reflection explicit) in artistic learning, along with the role of production and perception. We believe that reflection about one's artistic abilities is a virtue, that it frees rather than hampers the artist, and that it can be shown to improve the artist's productive and perceptual abilities. We also believe that reflection should be considered along with perception and production in any assessment of the student's learning and progress in art.

Our goal is to build upon the naturally occurring reflective activities that have been observed in a variety of classrooms—the informal discussions that take place between teacher and student about the revision of a work, peer discussions of works, and students' thoughts about preliminary sketches and revisions, which, though usually discarded, can provide crucial information about the growth of a project from conception to completion.

Although this study is in its infancy, we have already begun to find that the process of making reflection explicit has helped students make decisions about their work. In one instance, a student who was dissatisfied with his work was helped to talk through the differences he saw between his first sketch and a second draft. During the course of this reflective activity, he came to see a new possibility to explore that he felt would resolve his feeling of dissatisfaction. We feel that a focus on reflection as well as on production and perception may provide a view into the learning process in the arts and may also provide a means of assessing progress and stimulating further learning.

The Development of Production, Perception, and Reflection in Children Gifted in the Arts

We have recently begun to look at children with exceptional abilities in the arts. It is clear that the difference between gifted and nongifted children is most readily apparent in the realm of production. In fact, it seems as

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though this difference is demonstrated very early on (Clark & Winner, 1985; Gardner, 1980). Longitudinal analyses of drawings produced by children who retained a strong interest in drawing in late adolescence has shown that the early drawings of such children are marked by advanced technical skill, more articulated contours, and far less stereotyped forms than the drawings of their less gifted peers. In addition, gifted children demonstrate strong and preferred themes, such as a fascination with linearity, and the majority of their drawings are repeated explorations of this theme.

We have begun to examine the extent to which the exceptional production skills of gifted children are also apparent in perception and reflection. In a study of perceptual skills in visually gifted children, we compared memory for visual information in gifted and nongifted children (Rosenblatt & Winner, 1988). We sought to examine incidental rather than deliberate memory in order to determine whether visually gifted children retain visual images without intentionally committing them to memory. Two groups of children were selected by their art teachers to participate: those who demonstrated unusual talent in drawing, and those with only average ability. The children (aged eleven to fourteen years) were shown pairs of pictures and were asked which one they preferred. We used this preference task as a means of ensuring that the children looked at the pictures carefully without knowing that their memory would be tested later. After a fixed period of time, the children were shown the pairs again; but this time, one picture in each pair was an altered version of that seen originally. The alterations varied systematically along dimensions of line quality, composition, color, form, and content. The children were asked first to identify the altered picture and then to describe the details of the changes they observed.

The gifted children were not only better able to identify the altered pictures, but they also recalled more of the specific details. In fact, these children scored even higher than a sample of adult artists tested during the piloting of the materials.

Surprisingly, 24 percent of the gifted children were left-handed (the incidence of left-handedness is about 10 percent in the population at large). In contrast, none of the nongifted children was left-handed. Thus, there is some indication that brain laterality may play a role in giftedness in visual art.

A second study addressed the question of whether the superior memory skills of children gifted in drawing are domain specific (limited to visual stimuli) or more general (cutting across visual and verbal stimuli) (Rosenblatt & Winner, unpublished manuscript). Tests of visual and verbal memory were constructed and administered to children gifted in drawing and those gifted in creative writing. These tests were designed to compare

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memory for stimuli functioning aesthetically (ones in which repleteness, expression, and composition are meant to be noticed) with memory for stimuli functioning nonaesthetically (ones in which aesthetic properties are irrelevant). The aesthetic pictures consisted of paintings, line drawings, and collages; the nonaesthetic pictures were maps, diagrams, and graphs. The verbal memory test was analogous to the visual test and also included items with an aesthetic function (poems, stories) and a nonaesthetic function (newspaper articles, instructions).

There were striking differences in memory skills between the two groups. Visually gifted children demonstrated superior recall for visual stimuli; verbally gifted children had superior recall for verbal stimuli. These results are consistent with our previous findings of domain-specificity in aesthetic perception with ordinary subjects (Winner et al., 1986). Perceptual skills within a domain of talent do *not* generalize to other domains, even in children with exceptional ability.

Differences were found in the way the visual and verbal children recalled pictures which functioned aesthetically vs. nonaesthetically. The visual children recalled nonaesthetic pictures as well as aesthetic ones. Verbal children recalled the aesthetic items better than the nonaesthetic. Thus, it appears as if for visually gifted children, all visual stimuli function aesthetically. All images are accorded the kind of "thick," sensuous attention ordinarily reserved for aesthetic objects. In contrast, nonvisual children respond to pictures identified as "art" with thick attention, but look right through the sensuous properties of those identified as nonart.

Both groups of children had heightened reflective skills. They were unusually aware of the effects they attempted to achieve and of the "phases" they passed through in the development of their work. For example, when discussing a painting she had done a year ago, an eleven-year-old girl stated that she had been fascinated with Oriental painting at that time and had tried to incorporate certain techniques into her own style. When she felt she had exhausted the use of these techniques, she decided to explore other possibilities. It is important to note that these children had received no training in reflection. Hence, reflection seems to be a skill that develops precociously in visually gifted children, based perhaps on their rich experience with the visual medium. The spontaneously greater development of reflective capacities reported here could be taken as evidence that reflection is appropriate to the visual arts, that it is natural and enriching, not destructive and constraining.

Conclusion

The work conducted over the years at Project Zero suggests that development in the arts involves not one unified course of growth, but rather

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three separate and distinct lines. Perception proceeds linearly and is not marked by stagelike properties. In contrast, production follows a U-shaped curve and is more clearly divided into highly characteristic stages. Like perception, reflection seems to progress linearly with age, but also mirrors the development of more general knowledge and conceptions of the world.

Our studies of giftedness suggest that children with talent in production may possess heightened skills in perception and reflection as well. Not only do they seem to *see* artworks (and even nonaesthetic visual stimuli) differently than do nongifted children, but they also appear to think and talk about them in a qualitatively different way. Undoubtedly nongifted children can be helped to perceive and reflect at a higher level were perception and reflection included in ordinary art education. For the most part, children learn to produce in art classes, but not to look at, reflect on, or talk about artworks. Perhaps it is because of the exclusive focus on production that involvement declines with age. If art education fostered perception and reflection as well, children at the conventional stage might retain an interest in the arts. That is, they might be stimulated to look and reflect even if they do naturally resist exploration and play in their own drawings.

REFERENCES

Arnheim, R. (1974). *Art and visual perception*. Berkeley: University of California Press.

Altschuler, P., & Hattwick, L. (1947). *Painting and personality: A study of young children*. Chicago: University of Chicago Press.

Carothers, T., & Gardner, H. (1979). When children's drawings become art: The emergence of aesthetic production and perception. *Developmental Psychology*, *15*(5), 570-580.

Clark, P., & Winner, E. (1985). Unpublished data, Harvard Project Zero.

Freeman, N. (1980). *Strategies of representation*. London: Academic Press.

Gardner, H. (1980). *Artful scribbles*. New York: Basic Books.

Gardner, H., Winner, E., & Kircher, M. (1975). Children's conceptions of the arts. *Journal of Aesthetic Education*, *9*(3), 60-77.

Gardner, H. (1970). Children's sensitivity to painting styles. *Child Development*, *41*, 813-821.

Golomb, C. (1981). Representation and reality: The origins and determinants of young children's drawings. *Review of Research in Visual Arts Education*, *14*, 36-48.

Goodenough, F. L. (1926). *Measurement of intelligence by drawing*. New York: Work Book Co.

Goodman, N. (1976). *Languages of art*. Indianapolis: Hackett.

Goodnow, J. (1977). *Children's drawings*. Cambridge: Harvard University Press.

Piaget, J. (1963). *The origins of intelligence in children*. New York: Norton.

Piaget, J., & Inhelder, B. (1967). *The child's conception of space*. New York: Norton.

Rosenblatt, E., & Winner, E. (1988). Is superior visual memory a component of superior drawing ability? In L. Obler and D. Fein (Eds.), *The exceptional brain*. New York: Guilford Press.

Rosenstiel, A., Morison, P., Silverman, J., & Gardner, H. (1978). Critical judgment: A developmental study. *Journal of Aesthetic Education*, *12*(4), 95-107.

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193.63.235.17 on Fri, 30 Jan 2026 09:14:19 UTC
All use subject to https://about.jstor.org/terms

Schaefer-Simmern, H. (1948). *The unfolding of artistic activity*. Berkeley: University of California Press.

Wan, D., & Winner, E. (1980). Unpublished data, Harvard Project Zero.

Winner, E., & Gardner, H. (1981). The art in children's drawing. *Review of Visual Arts Research*, *14*, 91-105.

Winner, E., & Pariser, D. (1985). Giftedness in the visual arts. *Items*, *39*(4), 65-69.

Winner, E., & Rosenblatt, E. (unpublished). Cognitive skills and personality profiles of children gifted in the arts.

Winner, E., Blank, P., Massey, C., & Gardner, H. (1983). Children's sensitivity to aesthetic properties of line drawings. In *Proceedings of NATO conference on the acquisition of symbolic skills*. New York: Plenum Press.

Winner, E., Mendelsohn, E., Barron, J., and Gardner, H. (1980). Unpublished data, Harvard Project Zero.

Winner, E., Rosenblatt, E., Windmueller, G., Davidson, L., and Gardner, H. (1986). Children's perception of properties of the arts: Domain-specific or pan-artistic? *British Journal of Developmental Psychology*, *4*, 149-160.

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